



3C INTELLIGENCE

AI-Powered Urban Risk Engine & Interactive SaaS Dashboard

Project Goals

To build a reproducible data pipeline and unsupervised machine learning clustering model that partitions 50 municipal jurisdictions into distinct risk cohorts based on crime rates and population, providing policy-makers with an actionable risk assessment tool.

Key Challenges

Raw municipal demographics differed by magnitudes (millions vs units). Furthermore, standard clustering groups are mathematical and require severity-weighted labels without human intervention or bias.

Engineering Solutions

Applied Z-score normalization (StandardScaler) before fitting a K-Means model. Identified the optimal cluster count (K=3) via Elbow and Silhouette verification scans, and implemented a custom crime severity score index to map clusters to **Safe**, **Moderate Risk**, and **High-Risk** categories. The model and scales are serialized to disk for instantaneous, production-ready Flask API serving.

CORE TECH STACK
• Machine Learning: Python, Scikit-Learn, NumPy, Pandas
• Backend Framework: Flask API, Python Server, Pickle serialization
• Visualizations: Matplotlib, Seaborn Pipeline Plots
• Dashboard UI: HTML5, Custom Vanilla CSS Grid/Flexbox, JavaScript

PROJECT PERFORMANCE METRICS
50 CITIES ANALYZED
0.74 SILHOUETTE SCORE
K = 3 OPTIMAL CLUSTERS
<10ms API LATENCY

Visual Diagnostics & Business Outcomes

Below are key plots exported from our offline data science training pipeline, alongside aggregated statistical groupings that populate our real-time database.

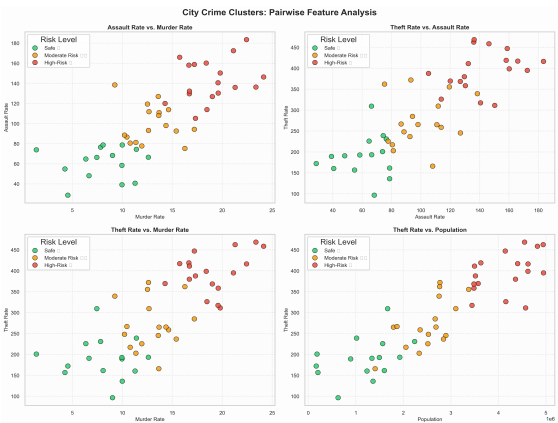


Figure 1: K-Means cluster separation showing clear risk cohort boundaries.

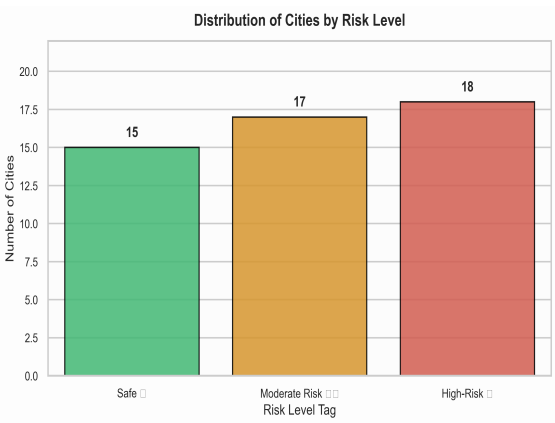


Figure 2: Distribution of jurisdictions inside the Safe, Moderate, and High-Risk categories.

Aggregated Risk Level Statistics:

Cohort	Cities	Avg Population	Avg Murder	Avg Assault	Avg Theft	Avg Crime Index
High-Risk	18	4,154,583	19.0	142.6	395.1	122.9
Moderate Risk	17	2,512,826	13.0	99.8	271.8	85.1
Safe	15	1,168,562	8.1	61.2	190.3	56.1

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Are you looking to integrate advanced machine learning models into interactive full-stack dashboards? I specialize in building end-to-end data analytics applications, custom prediction APIs, and beautiful responsive dashboards (Light/Dark themes). I guarantee high performance, pixel-perfect design, clean code, and production-ready deployable applications.

Contact me today to discuss your project requirements!